Local Union Organization and Law Making in the US Congress

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The political power of labor unions is a contentious issue in the social sciences. Departing from the dominant focus on membership size, we argue that unions' influence on national law making is based to an important degree on their local organization. We delineate the novel hypothesis that the horizontal concentration of union members within electoral districts matters. To test it, we draw on administrative records and map the membership size and concentration of local unions to districts of the US House of Representatives, 2003–12. We find that, controlling for membership size, representatives from districts with less concentrated unions have more liberal voting records than their peers. This concentration effect survives numerous district controls and relaxing OLS assumptions. While surprising for several theoretical perspectives, it is consistent with theories based on social incentives. These results have implications for our broader understanding of political representation and the role of groups in democratic politics.

The power of labor unions to influence elections and law making is a central concern in the study of democratic representation. Research in political science, economics, and sociology has commonly conceived of union power in the political arena as a question of aggregate group size, assuming that more members bring more votes and, perhaps, money. For instance, scholars have examined the effect of unionization on legislative voting in the US Congress (Box-Steffensmeier, Arnold, and Zorn 1997; Freeman and Medoff 1984; Kau and Rubin 1978; Seltzer 1995). Related research examines the macro-level relationship between unionization and outcomes such as turnout, economic policy, political equality, or poverty (Bartels 2008; Brady, Baker, and Finnigan 2013; Flavin 2016; Leighley and Nagler 2007; Radcliff and Davis 2000). This research views unions mainly through the lens of aggregate membership numbers, ignoring other organizational features. A recent review concludes that existing studies "commonly sum together membership over many different unions with the implicit assumption that organizational characteristics do not matter" (Southworth and Stepan-Norris 2009, 310).

In this paper, we argue that focusing solely on membership size provides an incomplete account of the organizational basis of union political power, and it significantly limits our broader understanding of political representation. Specifically, we develop and empirically assess the argument that the influence of unions on national lawmakers has significant roots in their local organization. We propose a novel hypothesis about the link between district-level union structure and law making. It states that the horizontal distribution of union members across local units within a congressional district—a feature we call concentration—shapes legislative voting: representatives elected in districts where union members are relatively concentrated should be less supportive of union positions than representatives elected in districts where an equal number of union members is dispersed across several unions. Theoretically, the hypothesis that membership concentration reduces union political power is controversial. To motivate it, we draw on foundational theories of collective action that point to the importance of selective social incentives in groups for political action (Olson 1965) and seminal

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behavioral research on the political significance of social interactions in local unions (Berelson, Lazarsfeld, and McPhee 1954). Unions have a federal structure with numerous constituent units at the local (i.e., establishment) level. Local unions lie at the heart of organized labor in the United States (Freeman and Medoff 1984, 34; also see Olson 1965, 66–76). They form the organizational base of the union pyramid, and this is where workers interact with each other on a regular basis, in the workplace and after work. At the local level, social incentives shape to what degree political resources of union members and their networks—votes as well as time and money—can be effectively mobilized to influence national law making. Because social incentives are more effective when groups are not too large, this logic suggests that unions' political influence in a district is higher when union members are distributed across many local units than when they are concentrated in few large units.[1]

To empirically evaluate the concentration hypothesis, we draw on extensive and geographically fine-grained administrative records from the Department of Labor. This largely neglected data source allows us to precisely map union membership size and membership concentration to electoral districts of the House of Representatives between 2003 and 2012. Strikingly, the data show that the concentration of union membership is orthogonal to the number of union members in a district. In line with previous research, we find that district-level union membership is significantly linked to legislative voting. However, we also find that district-level union concentration is an important determinant of legislative votes. After accounting for membership levels, legislators from districts with a relatively low union concentration have a substantively more liberal legislative ideology and a higher propensity to support the union position on individual key votes (e.g., health care) than those from high concentration districts. This concentration effect holds after accounting for state effects, period effects, flexible time trends, and numerous district-level characteristics. Importantly, our empirical strategy accounts for district-level economic concentration, such as the number of firms and employment concentration, as well as campaign contributions from business. This rules out the possibility that the apparent effect of union concentration is explained by economic concentration. The results are also robust when relaxing functional form modeling assumptions. In addition, we provide some historical evidence to further clarify that economic structure alone does not explain concentration. Exploring channels of influence, we find that concentration is linked to campaign contributions and the election of Democratic representatives.

Taken together, our analysis demonstrates that the concentration dimension of local union organization substantively shapes law making in Congress. These findings matter for our understanding of the relevance of unions for democratic representation as well as theories of groups in democratic politics more broadly. While union membership in the United States and several other countries has receded far below its postwar apex (Rosenfeld 2014), union members still constitute one of the largest organized groups in the political arena. Our results suggest that their local organization affects the making of national laws on a broad range of policies, with implications for large segments of the population and political inequality. Theoretically, the concentration effect we find is difficult to reconcile with the view, expressed for instance in the seminal work of Key (1964), that organizational fragmentation necessarily undermines the ability of groups to overcome collective action problems in politics. Instead, it is consistent with theories of political action in groups that account for social incentives. Our focus on local organization complements recent studies emphasizing that the political influence of unions is conditioned by the institutional environment, such as electoral rules and labor laws, or leadership (Ahlquist and Levy 2013; Anzia 2011; Anzia and Moe 2015; Flavin and Hartney 2015; Kim and Margalit 2016). It stands to reason that membership concentration may also matter for the political significance of other groups.

Another contribution of our paper lies in its empirical strategy. It departs from the heavy reliance on survey data in quantitative research on unions. This enables us to overcome two important problems (Southworth and Stepan-Norris 2009). First, a fundamental drawback of mass surveys is that they typically do not provide detailed information on the local union to which a member belongs. This precludes the possibility of examining the concentration of union membership. The second limitation concerns the counting of union members in a particular locality. State-level estimates of union density from the Current Population Survey (Hirsch, Macpherson, and Vroman 2001) are used frequently across the social sciences. However, the number of survey respondents is too small to provide membership numbers for electoral districts. Hence measurement is bound to be quite noisy and may be upward biased (Southworth and Stepan-Norris 2009). Many previous studies therefore rely on state-level measures only; others limit their coverage to metropolitan statistical areas (Box-Steffensmeier et al. 1997). In this paper, we use mandatory reports (so-called

1. Taking a different perspective, comparative political economists have studied the centralization of union organizations for wage bargaining as well as broader corporatist arrangements (e.g., Iversen 1999; Pontusson, Rueda, and Way 2002). Our approach is complementary to this important line of research, which does not examine the political effects of district-level union organization.

LM forms) filed by local unions to the Department of Labor. Their submission is a legal requirement for most unions, nonsubmission and incorrect submissions are penalized, and the Department of Labor conducts regular audits. We have retrieved all available raw data for around 30,000 individual unions (from 2000 to 2012) from the Department of Labor and processed them such that we are able to construct annual measures of union membership and membership concentration by congressional district. The resulting measures provide a new empirical perspective on the structure of organized labor in the twenty-first century; they may also be used to address a variety of questions not considered in this paper.[2]

LOCAL ORGANIZATION AND POLITICAL INFLUENCE

We proceed to lay out the theoretical perspectives that motivate our subsequent empirical sections. Going beyond the common conception of membership size as the essential power resource, but echoing several strands of research, we advance the argument that organized labor's effect on national legislators is strongly tied to the organization of local unions in an electoral district.

Following Olson (1965, 136) and many others, labor unions are viewed as groups that have been organized to serve the economic interests of their members relative to their employers, through collective bargaining over wages and benefits. Unions may turn to political action to pursue policy preferences and ideas of their members and/or leaders, though this does not happen by default. We contend that local organization shapes to what degree political resources of union members and their social networks can be effectively mobilized to shape policy.

The concentration hypothesis

In a polarized two-party system, one party will be closer to the political leanings of most union members. Since the New Deal and especially early postwar years, that role has been played by the Democratic Party (Dark 1999; Lichtenstein 2013; Schlozman 2015). Democratic representatives in Congress are predictably more supportive of policies favored by unions than their Republican counterparts (Box-Steffensmeier et al. 1997; Seltzer 1995).[3] A majority of union members regularly reports favoring Democratic over Republican candidates (Rosenfeld 2014, 176). And while political economy models highlight that unions' narrow economic interests may diverge due to their specific occupational or sectoral structure, issue bundling inherent in two-party competition nonetheless produces a fairly stable alignment between Democrats and labor unions. Thus, the main problem of organized labor in the political arena is to get Democratic lawmakers elected to protect or expand liberal economic policies in Congress. This requires votes, money, and other resources to win office.[4]

We argue that local union organization matters for achieving this goal far more than has previously been recognized, and we focus on two important dimensions: union membership and the hitherto ignored degree of union concentration. By concentration we refer to the degree to which union members in a congressional district are concentrated in few local unions versus being distributed more evenly across several locals. It is usually argued that a higher number of union members entails more political influence, since it implies a larger pool of votes and other resources (Box-Steffensmeier et al. 1997; Masters and Delaney 2005, 369; Olson 1965, 68). We do not dispute this hypothesis. Everything else equal, increasing union membership brings more political clout. However, our point is that the usual focus on membership counts or union density leaves out a politically significant feature of union organization.[5]

The significance of local unions has already been discussed in a less widely cited part of Olson's (1965) seminal analysis of collective action. He points out that the federal structure of organized labor, with thousands of local unions as the basic organizational unit, may be conducive to collective action in a group that is large in the aggregate. Even in the absence of external enforcement, social interactions can sustain individual contributions toward the collective good—as long as local unions are not too large (Olson 1965, 66–97). This setting fosters social interactions between local union members and entails selective incentives to engage in costly activity on behalf of the group.[6]

2. Scholars have noticed the large potential of the Department of Labor's LM forms for social science research, though it appears that "the cost in synthesizing a large sample has thus far deterred systematic analysis" (Southworth and Stepan-Norris 2009, 312). There have been studies using smaller subsets of these records (Martin 2008; Zullo 2008). We provide the first comprehensive analysis for a multiyear period covering the whole country.

3. Excluding Southern Democrats.

4. More generally, rational theories of electoral mobilization assume that turnout matters for policy mostly or exclusively by influencing which ideological type of politician wins office (e.g., Abrams et al. 2011). Research on legislative voting in Congress also suggests that voters affect law making more by influencing who wins the election than by changing the position of elected politicians (Bartels 2008; Lee, Moretti, and Butler 2004).

5. This is a ceteris paribus argument. We are not arguing that unions' political influence is exclusively due to local interactions. National unions and the competency and ideology of their leaders are important (e.g., see Ahlquist and Levy 2013; Dark 1999).

6. While Olson (1965) recognizes the potential importance of social incentives for unions, he presumed that most local unions had become too large for this mechanism to be effective. However, our descriptive data show otherwise for 2003–12. While there are some very large local unions, the

Going beyond Olson (1965), we take this logic to apply to political behavior on behalf of the group (i.e., union) more broadly. This includes voting as well as contributing money or time to a political campaign. It may also include efforts to foster norms of solidarity or to shape policy preferences in line with group ideology (Ahlquist, Clayton, and Levi 2014; Kim and Margalit 2016). The basic mechanism is that the prospect of approval, respect, or companionship being awarded or withdrawn alters the individual calculus of political action. For example, Abrams, Iversen, and Soskice (2011) present a formal model that elucidates how networks of family, friends, or coworkers shape participation in large elections (the same logic applies to political contributions). People vote despite a negligible probability of being pivotal if their social network attaches enough importance to voting for a particular party or candidate. Voting consistent with the group norm is rewarded while deviations are punished, and in equilibrium voting and punishment are self-enforcing. Social pressure remains a potent motive for political behavior in the internet era (Druckman and Green 2013). It does not require explicit action by group leaders, even though they may try to harness it.

Of course, empirical research started to document the relevance of social interactions in unions for politics well before the publication of Olson's (1965) theory. In their study of the 1948 presidential election, Berelson et al. conclude that social interactions within the same unionized plant were driving higher support for the Democratic party in a context where mobilization by local union leaders was low (Berelson et al. 1954, 37–53). Since then, a large body of research has confirmed the relevance of social incentives for political behavior more broadly, increasingly drawing on field experiments, though usually not considering unions (e.g., Druckman and Green 2013; Gerber, Green, and Larimer 2008).

Drawing on the preceding arguments, we propose that both the total number of union members and the horizontal concentration of members across different local unions within the same electoral district matter for democratic representation. It is uncontroversial to claim that a higher number of union members in a district should be linked to more consistent legislative support for proworker policies by elected lawmakers. Our novel hypothesis is that the concentration of members also matters: everything else equal, lawmakers from a district with highly concentrated union membership should be less supportive of pro-union legislation than lawmakers from districts where the same membership is distributed across more unions. The concentration hypothesis follows from the diminished efficacy of social incentives when many members are concentrated in one or few large unions.

Two theoretical qualifications are in order. First, concentration will not matter politically where union membership is so small that not even a fully mobilized membership is politically relevant. This threshold may vary depending on district characteristics (such as competitiveness). In the empirical section, we explore this issue in an interactive model. Second, the social interaction logic implies that high membership concentration undermines social incentives. It does not necessarily suggest that more membership fragmentation is always better. As unions become very small, approaching the (hypothetical) extreme of a one-member "group," social pressure will not exist at all.[7] Our empirical concentration measure will reflect this consideration.

Countervailing forces

Other approaches in the literature point to countervailing mechanisms that push in the opposition direction of the concentration hypothesis. One view is that collective action problems between the leaders of different local unions undermine the electoral mobilization of union workers (e.g., see Key 1964, 66). With numerous unions in the same electoral district, union leaders deciding how much effort to put into political action are more likely to face a collective action problem in which they have incentives to free ride on the political efforts of others. As a result, the dispersion of union members across multiple unions in a district should reduce, rather than increase, the political influence of organized labor. Thus, one may plausibly expect that, controlling for union membership, higher union concentration leads to more influence through the selection of more union-friendly legislators.[8] This is a clear-cut argument, consistent also with the standard model of Olson (1965). That said, there are theoretical reasons to suspect that the importance of collective action problems among local leaders is diminished by the same mechanisms that underlie the concentration hypothesis. Social incentives can encourage local leaders to do their part despite the temptation to free ride. Less concentrated unions increase members' ability to hold their leaders to account, thus inducing them to political action via social pressure.

Another argument is that higher union concentration should strengthen the political power of unions if concentration goes hand in hand with more homogeneous policy preferences.

typical local union is small: it has slightly more than 100 members (see appendix A.2, fig. A.2.1; see also Freeman and Medoff 1984, 34–35).

7. This point is highlighted in the formal model of Abrams et al. (2011, 244).

8. A related argument is that organizational fragmentation may reduce labor's political clout through inefficient allocation of political resources (Lichtenstein 2013, 143–44).

Unions' narrow policy preferences may be based on occupation, industry, or firm interests. Higher union concentration may then imply that policy preferences are more homogeneous, making it easier perhaps to overcome collective action problems and increasing policy makers' incentives to be responsive (Busch and Reinhardt 2000). Prior research suggests that some heterogeneity in the narrow material interests of unions does not necessarily prevent joint political action. One point is that the dynamics of two-party competition leads to policy bundling along a single-dimension of politics despite a potentially large multidimensional policy space; it usually means that different unions side with the same political party on a broad range of issues (Rosenfeld 2014; Schlozman 2015). Unions have also constructed broader "communities of fate" that cut across narrow economic interests and sometimes take costly actions on behalf of other or broader groups (Ahlquist and Levy 2013; Lichtenstein 2013). Moreover, unions interested in their long-term survival have incentives to organize across industries (Kremer and Olken 2009), which implies that membership concentration need not reflect preference homogeneity.

This discussion highlights the fact that the effect of membership concentration on lawmakers is theoretically ambiguous. Only if the force of social mechanisms is sufficiently strong should we see that lower concentration leads to more legislative support of policies favored by unions and their members. In that sense, the concentration hypothesis is not "self-evident." It merits careful empirical testing.[9]

MAPPING ORGANIZED LABOR

We map our two central features of local union organizations—their size and concentration—to electoral districts for the US House of Representatives during the 109th–112th Congress. We use administrative data covering almost 30,000 local unions for more than a decade, based on more than 300,000 individual reports. Our data set provides a new empirical perspective on the structure of organized labor in the United States, and it allows for a reexamination of the political influence of labor unions on law making. In this section, we describe our measurement strategy and data.[10]

Using LM forms

We analyze and aggregate mandatory reports filed annually by local unions with the US Department of Labor. As this is not a widely used data source in political science, some exposition of its advantages and drawbacks will be helpful. The legal basis for these reports is the Labor-Management Reporting and Disclosure Act (LMRDA) of 1959, which started as a movement for greater union democracy (initiated by the American Civil Liberties Union) but was transformed by legislators into a push to curtail the economic and organizational power of unions (Aaron 1960).[11] The act introduced a comprehensive system of reporting: unions have to file an initial report with the Office of Labor-Management Standards (OLMS) followed by a yearly report using a so-called LM form. While the level of reporting detail varies by union income, all yearly forms include information on the number of union members and the address of the union office. For the public sector, the Civil Service Reform Act (CSRA) of 1978 affirmed union rights in the public sector created by previous presidential executive orders (Coleman 1980, 202). It also created a system of reporting similar to the one in place for the private sector (also overseen by OLMS).

Their legal basis makes LM forms a reliable source of information on unions and their members. Filing LM forms is mandated by law and failure to report, or reporting falsified information, is made a criminal offense in the LMRDA punishable by a fine of up to $100,000 and/or imprisonment of up to one year.[12] As a further incentive against misreporting, the validity of union reports is continuously verified. Under its Compliance Audit Program, OLMS selects a number of unions for detailed audit each year.[13]

Using LM forms has important advantages over using measures derived from surveys. Multipurpose surveys usually ask only about union membership without any reference to the particular union in question. This precludes studying the organizational basis of unions and probably explains the focus on membership size in the literature on the political power of unions. Even with respect to union membership there are well-known measurement problems. For example, the Current Population Survey (the most widely used source for union membership) uses a rather broad question wording, querying

9. As discussed in an earlier note, a strong literature in comparative political economy studies the economy-wide centralization or coordination of wage bargaining (e.g., see Iversen 1999; Pontusson et al. 2002). Its focus on the vertical distribution of bargaining authority within union confederations is orthogonal to the focus of this paper on the horizontal concentration of union members at the level of electoral districts.

10. Our focus is on the House of Representatives rather than the Senate because this allows us to exploit within-state variation in union organization. This is especially useful because the organizational features move slowly during our period of observation.

11. While the ACLU has pushed for greater internal union democracy since 1947, LMRDA legislation occurred in the wake of the Select Committee on Improper Activities in the Labor Management Field hearings chaired by Senator McClellan from 1957 to 1959. For a summary of LMRDA's legislative history, see Aaron (1960).

12. The corresponding figure for unions covered by CSRA is $250,000 and five years imprisonment.

13. Unions were chosen based on random selection, OLMS's internal discretionary criteria, or based on complaints of union members.

respondents if the person in question is a "member of a labor union or of an employee association similar to a union." This invites overreporting leading to an overestimate of union membership (Southworth and Stepan-Norris 2009, 311). In addition, some systematic misclassification by respondents has been documented (Card 1996). A related issue is unit nonresponse, which plagues every survey.[14]

There are, of course, disadvantages to using administrative data. Perhaps the main potential drawback when using LM forms is that some unions are exempt from filing requirements. While each and every private sector union has an associated LM form, some public sector unions are not covered by the relevant laws and are not required to file regular reports. All unions representing postal or federal employees are covered. But unions that exclusively represent state, county, or municipal government employees are exempt. However, note the strong definition of exclusivity here. Unions almost exclusively representing, say, municipal government employees nonetheless do have to file if only one of their members is employed in the private sector or by the federal government. For instance, a police union that also includes private security employees will have to file a report. The same holds for a union of municipality workers that includes postal staff. Since the latter part of the twentieth century unions are no longer purely organized along craft and industry lines, but enroll workers of different sectors and occupations (Lichtenstein 2013, 249): "the UAW [United Automobile Workers] has recruited health-insurance clerks and prison guards, and its largest local west of the Mississippi represents teaching assistants, tutors, and readers at the University of California's ten campuses. Meanwhile, the United Steelworkers have organized Pittsburgh grocery workers, and the Communication Workers of America negotiate for state and municipal employees." Similarly, the American Federation of State, County, and Municipality Employees also organizes private sector workers in education, health, and home care. This behavior is consistent with evolutionary models of unions (Kremer and Olken 2009). Altogether, this suggests that the degree of noncoverage in our data is probably limited. Validating our data against governmental aggregated sources, we indeed find a very high degree of coverage (see below and appendix A.1; appendix available online).

Measuring membership and concentration

From the Department of Labor we obtain a database of digitized LM forms. The process of turning these raw data into measures of union organization for each House district consists of three steps: cleaning the data from entry errors, spelling mistakes, and so forth; geocoding; and district-level aggregation using the relevant quantities of interest.[15]

Using cleaned addresses, we produce a geolocation for each union address by processing each address via natural language processing and then matching it geographically.[16] Since not all addresses are necessarily complete, we use exact matching with attribute relaxation. This means that while exact matches are preferred, they can be produced at decreasing levels of precision: (1) the most precise geolocation is produced by an exact match to a segment of a street based on street and house or building numbers; (2) at the second level, matches are based on the ZIP portion of an address; (3) if this fails, matches are based on the city or on county regions; (4) if a whole address cannot be resolved, matches are based on the state portion of an address. However, the latter three strategies are rarely needed. More than 99% of the 358,051 union forms filed between 2000 and 2013 are matched based on either an exact address or a ZIP code. Given a union's unique geolocation, it is placed into the map of districts for the House of Representatives for each Congress using the cartography shapefiles for congressional districts from the Census Bureau.

With the geolocated union data in hand, we calculate two main measures of local union organization. First, the total number of union members in the district is the sum of members in all local unions. Given census-based congressional apportionment, population size is roughly constant across districts (and often identical within states), except in the seven at-large districts. Hence, it makes little difference whether we simply look at membership counts or normalize them by population. As discussed above, membership size is the dominant measure of union political power in the social science literature. Compared to surveys, our LM-based measure does not suffer from small sample problems and is geographically more accurate than those used in previous research (see, e.g., Box-Steffensmeier et al. 1997).

14. While the CPS has very low nonresponse rates, they (in line with the general trend) have risen from about 7% in 1995 to 11% in 2014 (Robison and Grieves 2014, 1339), increasing the need to rely on weighting procedures to produce national or state estimates of union membership.

15. We conducted checks of the Department of Labor's digitization of LM forms and find a high level of accuracy. For a randomly selected sample of 2,291 LM forms, we verified the coding of their content against the original documents. In 96.8% of all cases, digitized information on address and number of union members exactly match the original form submitted by the union officer. Most issues with the remaining 3.2% of deviating forms were minor and concerned mostly address details, such as small typographical errors. A number of those deviations result from mistaken entries by union officers, for example, by entering street addresses into the field intended for PO boxes (and vice versa). We address these issues first through data cleaning of all 358,051 LM form address fields, and second through a flexible natural language processing step preceding our geocoding procedure.

16. We use Texas A&M University's geocoding services; see Goldberg, Wilson, and Knoblock (2007) for details.

Second, following our theoretical discussion we calculate the degree of concentration versus dispersion of union members in an electoral district. Our main measure of union concentration is the four-union concentration ratio (CR4). It captures the share of all union members in a district who belong to the largest four unions. It is thus bounded between 0 and 1. In the case of perfect concentration, all members are concentrated in the top four unions. As membership becomes more dispersed, the ratio declines. Empirically, at the minimum the largest four unions capture only 14% of all members. The concentration ratio is intuitive and straightforward to interpret. Another desirable property is that it is not mechanically related to the overall union membership in a district. Group size and concentration may still be correlated empirically, but this will not be by construction. While the four-largest-unions threshold is somewhat arbitrary, it captures a large amount of variation in the data; it also follows a long tradition in the empirical analysis of firm concentration (Curry and George 1983), and it seems natural to apply the same criteria to unions. Alternatively, we can calculate the effective number of unions (analogous to the well-known effective number of parties measure), which turns out to be highly correlated with the concentration ratio ($r > 0.9$). All results reported below also obtain when using it.[17] Both measures capture the theoretical intuition that social pressure works best when membership is not too concentrated. This logic does not suggest that more fragmentation in the tail is always beneficial. In appendix A.4, we provide evidence that union concentration is rooted in historical organizational preferences.

Before describing the structure of local union organization, we report how we validated our data. While there is no "gold standard" of accurate union membership numbers, we compare our data to the widely used CPS-based measure of state-level union density. The two measures agree to a large extent (their correlation, averaged over all years, is 0.86). On average unionization levels based on estimates from the CPS are 1.9 percentage points higher than counts of members from LM forms. This difference is consistent with some degree of overreporting, induced by CPS's broad question wording (Southworth and Stepan-Norris 2009, 311). It can also be interpreted as an upper bound for the noncoverage of some public sector unions in our data, confirming that LM forms provide a rather comprehensive accounting of unions. For a more detailed discussion, see appendix A.1.

17. Appendix A.7 reports results for the effective number of unions; it also constructs a one-dimensional measure of union organization based on the ratio of union concentration to union membership suggested by a reviewer.

Two dimensions of union organization

A cross-sectional snapshot of union membership numbers (as population share) and membership concentration (CR4) across the map of congressional districts for the 109th–112th Congress is shown in figure 1. Panel A highlights that there is considerable variation in union membership even within states. For instance, states with relatively high average union membership, such as California or New York, show substantial geographic variation of union membership; even states with low average union membership, such as Florida or Texas, show pockets of high union density. Panel B shows that there is also considerable variation in union concentration among districts.

We have argued theoretically that membership numbers and concentration are separate dimensions of union organization. A district with a large number of union members might see these concentrated in a few unions or dispersed among many. Empirically, these two characteristics turn out to vary independently as well. In panel A of figure 2, we plot membership concentration against the logged number of union members for each of the 435 districts. The absence of any clear correlation ($r = -0.05$) is apparent from the widely scattered observations. This striking pattern suggests that we can, for the sake of exposition, distinguish between four combinations of membership size and concentration (high-high, high-low, low-high, low-low). Panel B plots these combinations in the map of House districts. While this simplified classification understates variability, one can see that there is variation within states that is not captured by membership counts. For instance, while California's fourth and twelfth congressional districts both contain an almost identical number of union members (about 40,000 during the 110th Congress), they vary dramatically in their level of concentration. In the fourth district, the four largest unions capture 92% of all members while capturing only about 50% in the twelfth. In a similar vein, Texas's seventh and eleventh congressional districts feature an equally low number of union members (about 2,500), but members are dispersed in the former (with a concentration ratio of 55%) and concentrated in the latter (97%).

What explains concentration?

Figures 1 and 2 raise the question of what explains variation in union organization. While there is a well-developed literature on the determinants of union membership (for a review, see Wallerstein and Western 2000), we are aware of no prior study that explains union concentration. In appendix A.4, we explore this question in more detail. To summarize, we suggest that district-level union concentration is partly rooted in history, shaped by the interaction of eco-

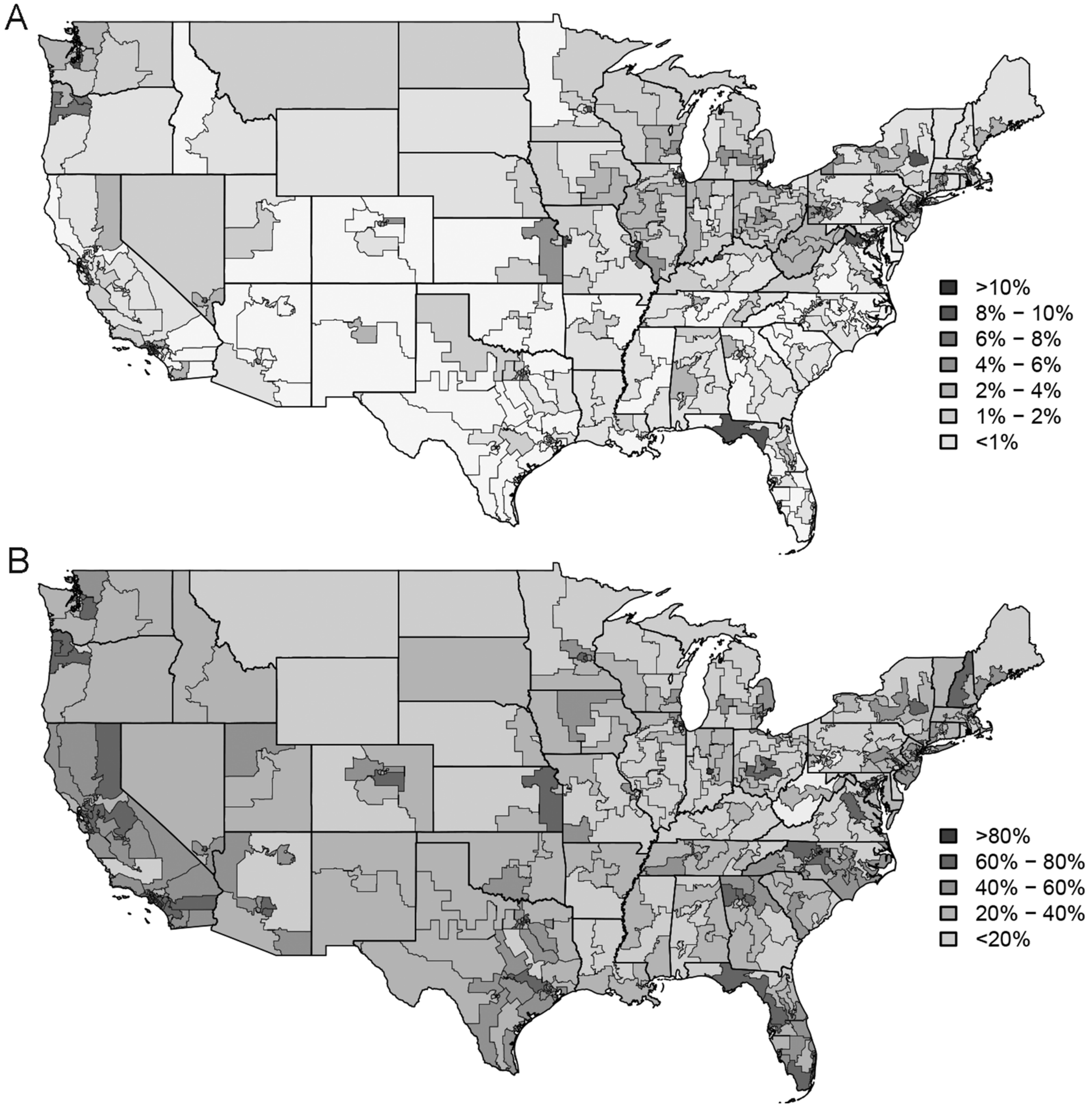

Figure 1. Union membership and concentration in the United States. The map shows average union membership and union concentration for each district of the 109th to 112th Congress. Entries are averages over time. Panel A shows union members as percentage of the total population; panel B shows union concentration (measured as four-union concentration ratio).

nomic structure and historical preferences for how unions should be organized. While the distribution of economic establishments puts a constraint on union concentration, their relatively large number (see appendix A.2) leaves a considerable amount of slack to be shaped by organizational preferences that, once established, are sticky. Using survey data from the New Deal era, a turning point in the history of organized labor (Lichtenstein 2013), we find that public preferences on how unions should be organized—narrowly along craft lines or more broadly—are clearly distinct from support for union rights, that is, whether unions should exist at all (appendix A.4, fig. A.4.1; figs. A1–A4 available online). Furthermore, in a regression analysis at the level of congressional districts we find that the contemporaneous link between economic concentration and union concentration is significantly moderated by preferences concerning union structure from the 1930s (ta-

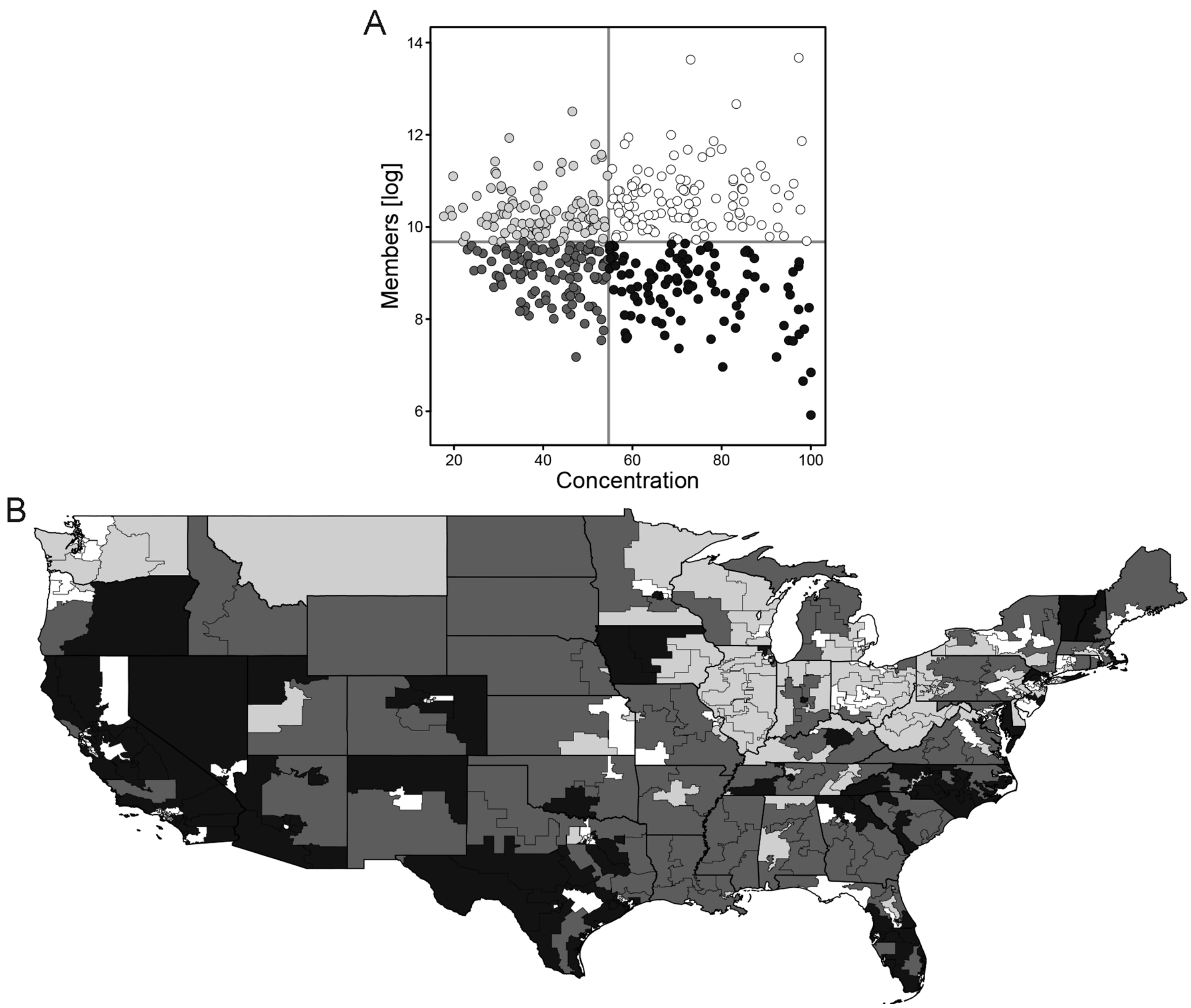

Figure 2. Relationship between union membership and concentration. This figure illustrates the independent variation of membership and concentration during the 109th to 112th Congress. Panel A plots CR4 concentration against (logged) membership in each district. Different shades of gray and black mark the four quadrants obtained by splitting both measures at the median. Panel B maps the distribution of the four membership-concentration combinations. All entries are averages for 2005–12.

ble A.4.1). These findings are broadly consistent with an organizational perspective of union behavior (Ahlquist and Levy 2013), which suggests that members of successful unions embrace the larger organizational goals or strategies set by (founding) leaders. They clarify that significant variation in union concentration can be traced back to preferences that are not hardwired into the current economic structure.

LOCAL UNIONS AND VOTING IN CONGRESS

Does the organizational structure of unions in a congressional district influence how the district's representative votes in Congress? Our data set allows us to directly assess our hypothesis that the political clout of organized labor is not only shaped by the number of union members (as found by previous studies) but also by the distribution of members across local unions.

Empirical strategy

We estimate a series of models for legislators' voting behavior as a function of local union characteristics and suitable controls. Our main model is:

$$y_{dst} = \mathbf{x}'_{dt}\boldsymbol{\beta} + \mathbf{u}'_{dt}\boldsymbol{\gamma} + \theta_s + \tau_t + \boldsymbol{\xi}_s\psi(t) + \epsilon_{dst}.$$

Our dependent variable is y_{dst}, a summary index of the voting behavior of the House representative from district d in state s during congressional term t based on a large number of votes.

We rely on the most popular measure, the first dimension of DW-NOMINATE scores calculated by Poole and Rosenthal (1997).[18] DW-NOMINATE scores measure legislators' revealed ideology on the dominant left-right dimension, where -0.75 is the most liberal and $+1.36$ the most conservative position in our data, and are comparable over time. The unit of analysis is the individual legislator in a given term.[19] In addition, we report results for individual key votes where unions took an explicit position.

The vector $\mathbf{u}_{dt}$ captures our two main variables characterizing local union organization as calculated from LM forms: the logged number of union members and the four-union concentration ratio.[20] As we have seen, different theoretical perspectives suggest opposite predictions about the direction of the concentration effect; our argument implies that higher concentration leads to less responsive policy makers for a given level of unionization. To capture common time shocks, such as midterm versus presidential electoral cycles or shifts in national mood, our model allows for Congress-specific effects τ_t. All our specifications include a set of time-varying district-level controls, $\mathbf{x}_{dt}$, that follow the literature on roll call voting (e.g., McCarty, Poole, and Rosenthal 2006): median family income, racial composition (percentage white), and level of education (percentage with BA degree or higher). We also control for the share of a district's workforce employed in the service sector, as this sector has been more resistant to unionization (Freeman and Medoff 1984, chap. 13), the share of agricultural employment, and a district's degree of urbanization. To further assess the concern that union concentration is driven by economic concentration, we will estimate specifications including additional controls for districts' economic structure. Appendix A.2 provides descriptive statistics for all variables used in our analysis.

The impact of time-invariant state-level characteristics is captured by state fixed effects, θ_s. Most notably, "right-to-work" laws in some states allow workers in unionized companies to opt out of union membership and paying associated dues, and thus shape the mobilization capacity of organized labor.[21] Collective bargaining rights for public sector workers also vary between states (Flavin and Hartney 2015). However, the inclusion of state fixed effects alone leaves open the possibility that omitted time-varying confounders affect our results. While an observational study is never able to rule out this possibility completely, we can account for linear and nonlinear time trends in state-level unobservables by including state-specific functions of time in $\boldsymbol{\xi}_s\psi(t)$.[22] Finally, ϵ_{dst} are residuals. We employ standard errors robust to within-state heteroscedasticity and correlation (Cameron and Trivedi 2005, 834).[23]

Our analysis focuses on the 109th–112th Congress (2005–12). This reflects research design considerations and data constraints. Since we pool observations from different districts, we focus on a single apportionment period during which district borders remain constant (i.e., 108th–112th Congress based on the 2000 census), with the exception of several cases of court-ordered redistricting in Georgia and Texas (we ensure that our results hold when these are removed).[24]

Results

Table 1 presents results from six specifications. All include both state and Congress fixed effects, as well as basic district characteristics. Column 1 includes our measures of union membership (the logged number of members) and concentration (the district four-union concentration ratio). These results confirm previous findings linking union membership to more liberal congressional voting outcomes (e.g., Box-Steffensmeier et al. 1997; Freeman and Medoff 1984; Kau and Rubin 1978; Seltzer 1995). Turning to the concentration hypothesis, our results show that concentration matters for legislative outcomes in the direction suggested by arguments about the importance of social incentives for group political action: holding overall membership constant, a unit increase in local union concentration is related to a 0.37 unit increase in conservative legislative ideology of that district's representative. In specification (2) we add the interaction of both union

18. Note that an alternative measure, pro-union voting scores assembled by the AFL-CIO, produces comparable results (in terms of estimates and significance); see appendix A.6 for details.

19. Including special elections to fill vacancies, the total number of possible DW-NOMINATE observations is 1,782. The analysis covers 1,767 because the data set retrieved from Poole's website (www.voteview.com) does not include scores for 15 legislators with too few individual votes to produce reliable estimates.

20. We take the log as the distribution of members is right skewed. In appendix A.6 we show that an alternative measure based on the share of union members yields the same substantive (in terms of sign and significance) results.

21. Their influence is captured by state-specific constants, since right-to-work laws are constant in our estimation sample (it ends before the only two reforms, which occurred 2012 in Indiana and Michigan, had time to materialize).

22. We specify $\psi(t)$ as restricted cubic splines with three knots.

23. The short times series ($T = 4$) and the slow-moving nature of union organization imply that a within-district analysis is not informative, as district fixed effects cannot be distinguished from the variables of interest.

24. We exclude the 108th Congress because our district-level controls from the American Community Survey are not available. But note that including the 108th Congress when backward-interpolating missing control variables via state-specific linear and quadratic time trends does not change our results (see appendix A.6).

Table 1. Union Membership, Concentration, and Legislative Voting

	(1)	(2)	(3)	(4)	(5)[a]
Union members	−.075	−.084	−.073	−.072	−.073
	(.022)	(.019)	(.022)	(.023)	(.020)
Concentration	.368	.374	.369	.376	.311
	(.116)	(.122)	(.113)	(.118)	(.102)
Members × concentration		.044			
		(.098)			
State fixed effects	✓	✓	✓	✓	✓
Congress fixed effects	✓	✓	✓	✓	✓
District characteristics[b]	✓	✓	✓	✓	✓
District economic structure[c]			✓	✓	✓
State time trends[d]				✓	✓
N	1,767	1,767	1,767	1,767	1,767
R^2	.512	.496	.514	.533	.543

Note. 109th to 112th Congress. Standard errors robust to arbitrary within-state correlation and heteroscedasticity. Analytical standard errors are possibly sensitive to small or unevenly sized clusters. We show in appendix A.6 that our results hold when employing a cluster bootstrap or a wild cluster bootstrap (Cameron, Gelbach, and Miller 2008).

[a] Partially linear model with flexible structure of confounders estimated via post-double-selection LASSO to minimize omitted variable bias (Belloni et al. 2013, 2017). See appendix A.5 for details.

[b] Controls for income (median household income), race (percent white), education (percent with college or postcollege degree), employment (percent in service industries, and percent in agriculture), degree of urbanization.

[c] Controls for number of firms and dispersion of employment across sectors. Firm data are spatially matched to congressional districts from economic census data; see appendix A.3 for details.

[d] State-specific time trends via restricted cubic splines.

variables (each centered for easier interpretation) and do not find model-based evidence for a conditional relationship between membership and concentration. Given the empirical pattern evident in panel A of figure 2, this is not surprising.

An alternative explanation not fully ruled out by the first two specifications is that union concentration might simply be a mechanical consequence of industrial structure. This is a relevant endogeneity problem, as firm or occupational structure (or the unobservables predicting them) may well affect both union concentration and legislative outcomes. Our empirical exploration of the determinants of union concentration in appendix A.4 indicates that economic structure alone is not sufficient to explain concentration. However, addressing this endogeneity concern also requires controlling for suitable proxies of economic structure in the analysis of legislative voting. We include two additional variables. The first is the number of firms, which we calculate from the Census Bureau's Economic Census.[25] The second is the dispersion of employment over occupational sectors calculated from American Community Survey data on sectoral employment shares. Together, both measures capture key features of industrial structure. As can be seen from column 3, their inclusion does not appreciably change the estimated union effects. This strengthens our interpretation that union concentration is a distinct and relevant phenomenon and not just a mirror image of economic structure.

Our previous specifications focus on within-state variation between districts and rule out as confounders both time-invariant state characteristics (such as culture) and idiosyncrasies of a given Congress. In specification (4) we aim to move closer to ruling out some time-varying state confounders as well, by including flexible state-specific time trends. Thus, any remaining omitted state-level variable would have to exert a highly nonlinear influence on congressional voting to bias our results. In specification (5) we move to subject our analysis to an even stricter test by estimating a partially linear model using the post-double selection estimator (Belloni, Chernozhukov, and Hansen 2013, 2017). The key idea of this model is to account for confounders in a very flexible way by dropping the restriction of a linear effect of covariates and by allowing for arbitrary higher order interactions among covariates. This leads to hundreds of covariate terms, of which a subset is selected using a standard variable selection tool (the LASSO). Importantly, this approach goes beyond a causally naive model selection tool by jointly estimating three equations: the outcome equation, where legislative ideology is the dependent

25. See appendix A.3 for details.

variable, and selection ("treatment") equations, where the dependent variables are union concentration and membership. This explicitly accounts for the logic of omitted variable bias: confounders (and their respective transformations) that matter in the selection stage are kept in the model even if they have only moderate weight in the outcome equation. For a more detailed exposition, see appendix A.5.

In both extended specifications, we find that the coefficient of union membership changes very little. Accounting for flexible state-specific time trends slightly increases our estimate for union concentration, while it decreases somewhat in the final specification. However, both specifications confirm our hypothesis that there is a statistically and substantively significant link between the concentration of union membership and legislative outcomes. Substantively, our results imply that a standard deviation increase in (log) union membership is related to a 0.16 ($\pm$0.05) standard deviation increase in liberal legislative ideology, while a standard deviation decrease in concentration is related to a 0.14 ($\pm$0.04) standard deviation increase in liberal ideology. These are politically relevant magnitudes, and we demonstrate below that they are indeed sufficient to affect the outcomes of major votes.

Assessing proximate district confounders

Table 2 assesses how robust our results are to the inclusion of what we call proximate district-level confounders. These are political confounders that may in part also be the result of union organization. One may think that preexisting political preferences explain union size and concentration as well as why representatives are more left-leaning in some districts than in others. Several scholars argue that union strength is plausibly exogenous to political preferences, conditional on the macro context (i.e., union laws) and economic conditions, because the motivation to join or even form a union is mainly economic. For example, Olson (1965, 153–55) argues that union political action is a by-product of economic activism (see also Ahlquist and Levy 2013, 16). Nonetheless, it is prudent to consider the possibility that prior political preferences are an omitted variable. Otherwise one may remain worried that our results simply reflect district partisanship. To address this issue, we employ two proxies. Our first measure is the district-level Democratic vote share in the 2000 presidential election. Since election outcomes are partly a consequence of union political mobilization and perhaps even "preference activation" (Ahlquist and Levy 2013; Kim and Margalit 2016), we use the last presidential election prior to our period of analysis. Our second proxy is the average district-level political ideology of campaign donors, as inferred from their donation patterns by Bonica (2014). Including these variables means that our analysis partials out unions' prior impact on district preferences and election outcomes. Specifications (1) and (2) in table 2 show that adding both proxies does not affect the estimated effect of union concentration. The coefficient on union members is somewhat dampened, but it remains substantively important and significant at conventional levels.

Table 2. Proximate District Level Confounders

	(1)	(2)	(3)	(4)	(5)
Union members	−.059	−.045	−.065	−.076	−.032
	(.024)	(.022)	(.023)	(.021)	(.021)
Concentration	.399	.296	.387	.373	.318
	(.105)	(.102)	(.124)	(.117)	(.111)
Presidential vote share[a]	✓				✓
Donor ideology[a]		✓			✓
Public union members			✓		✓
Corporate contributions[b]				✓	✓
N	1,767	1,767	1,767	1,767	1,767
R^2	.549	.632	.515	.513	.637

Note. 109th to 112th Congress. All models include state and Congress fixed effects and district-level controls for income, race, education, employment in services and agriculture, number of firms, sectoral employment dispersion, and urbanization. Standard errors robust to arbitrary within-state correlation and heteroscedasticity.

[a] Presidential vote share in 2000 election. Donor ideology estimated by Bonica (2014) in the 108th Congress. Missing values due to redistricting or insufficient information filled in using stochastic mean imputation (comprising 5.7% and 0.2%, respectively).

[b] Total corporate contributions (cube-root transformed) to candidate calculated from data of the Center for Responsive Politics. Missing values (0.7%) filled in using stochastic mean imputation.

We also account for (logged) corporate contributions to each legislator. This allows us to assess the concern that the concentration effect merely mirrors the structure of firms or business interests. Again, our findings on union organization remain robust.[26] Another specification includes the (logged) number of public union members in a district. Recent research has stressed the particular characteristics of public unions and their political influence (e.g., Anzia and Moe 2015; Flavin and Hartney 2015; Moe 2006). Hence, one may ask whether our results are mostly driven by public unions or if the sign of the effect changes when excluding public unions. Since the LM data do not allow us to exactly identify all public unions and their members (recall the discussion above), we calculate a district's number of public union members from an indicator created by selecting likely public unions based on their name.[27] Even after accounting for the degree of public membership in a district, our results show rather little change in our two union variables.

In sum, these additional results rule out preexisting political preferences or partisan leanings, as well as corporate strength, as an alternative explanation. A number of further specification and robustness tests, reported in appendix A.6, show that our findings hold under various alternative measurements and modeling assumptions.

Contributions and partisan selection

In the following, we try to shed some light on potential channels of influence. Based on our argument, the two dimensions of union organization are linked to the selection of partisan politicians, with unions rallying around Democratic candidates. We expect lower levels of concentration to be associated with a higher probability of a Democratic victory. Similarly, lower concentration should be associated with more financial contributions by unions and union members to sitting members of Congress. Again, alternative arguments based on coordination problems or preference heterogeneity suggest, if anything, the opposite. Table 3 indicates the empirical relevance of union organization for both contributions and partisan selection, supporting the logic behind our main hypothesis. The first column presents results from a linear probability model where the dependent variable indicates if a legislator is affiliated with the Democratic Party. The model accounts for observable district characteristics, as well as state and Congress fixed effects. Clearly, districts with more union members and lower levels of concentration show a higher propensity of selecting a Democrat, all else equal. In the second column we study the influence of local union characteristics on financial contributions. The dependent variable is the amount of candidate contributions from labor.[28] We find that local union structure relates to candidate contributions: a unit increase in (logged) union membership in a district raises labor contributions by $11,500. Even more notably, a unit decrease in concentration of members leads to a $66,800 increase in labor contributions, holding everything else constant.

Table 3. Candidate Selection and Labor Contributions

	Democrat[a]	Labor Contributions[b]
Union members	.068	.115
	(.019)	(.038)
Concentration	−.319	−.668
	(.114)	(.245)
N	1,765	1,767
R^2	.430	.309

Note. Models include state and Congress fixed effects and district-level controls (see table 2). Standard errors robust to arbitrary within-state correlation and heteroscedasticity.
[a] Linear probability model.
[b] Labor contributions are in 100,000s of dollars. Model estimated on cube-root transformed contributions, with estimated coefficients transformed back to dollar scale.

Union influence on key votes in the 111th Congress

We have shown that union characteristics matter for legislators' revealed ideology. So far, we have measured the former as the one-dimensional (unobserved) variable underlying representatives' (observed) voting record. We now turn to individual votes on key legislation where unions took explicit positions. In particular, we study all 35 major pieces of legislation in the 111th Congress that are part of the political scorecard of the AFL-CIO and code whether a representative votes in line with the union recommendation. These votes include, among others, the final passage of the Affordable Care Act (H.R. 3590) on March 21, 2010, and the Jobs for Main Street Act (H.R. 2847) on December 16, 2009. Health

26. We calculate corporate contributions similar to Carnes (2013) using the raw campaign finance contribution data from the Center for Responsive Politics. We sum Federal Election Commission–reported contributions to candidates from all "business" sectors (i.e., excluding labor and single-issue donations). Our count includes both individuals and PACs (but using either alone does not change our results).

27. In other words, we select unions that are likely to contain public employees by using regular expressions containing terms such as "firefighters," "police," "county," or "public" employees. While this does not, of course, yield a precise classification of public unions, it captures the degree to which (likely) public employees are present in a given district.

28. It is calculated in the same way as corporate contributions (see above), using contributions from the labor sector.

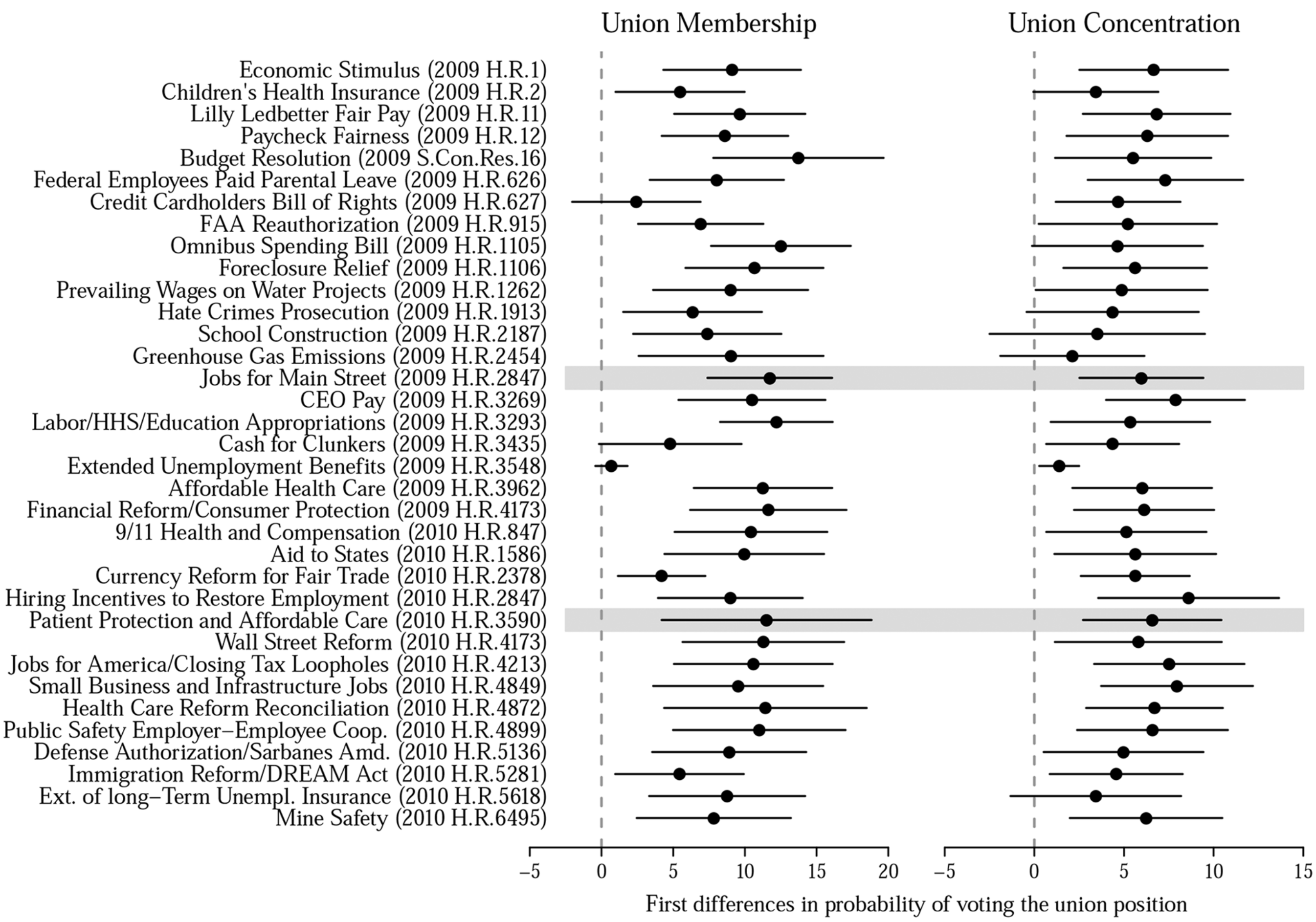

Figure 3. Unions and key votes in the 111th Congress. This figure shows the influence of union characteristics on key votes (as defined by AFL-CIO) in 2009 and 2010. Each row entry represents first differences in the probability of a representative casting a vote in line with the AFL-CIO-supported position as function of an SD increase in membership and an SD decrease in concentration. All models include district controls and state random effects and allow for arbitrary within-state correlation. Line segments are 95% confidence intervals.

care reform is widely considered one of the major reforms of the new century, something that unions had supported for decades but failed to achieve. Both votes were razor tight and highly partisan (219 vs. 211 and 217 vs. 212 in favor).

We estimate logit models for each key vote accounting for district characteristics (the same as in table 1) and state random effects with standard errors robust to within-state correlations.[29] Figure 3 plots first differences in the probability of a representative casting a vote in line with the AFL-CIO-supported position together with 95% confidence intervals.[30] It underscores the substantive effect of union organization on legislative behavior. In many of the votes that matter to the AFL-CIO, a standard deviation increase in membership increases the probability of a representative voting the pro-union position by almost 10 percentage points. The respective effect size for the influence of union concentration is around 5 points (all else equal). In the case of the final passage of the Affordable Care Act, a standard deviation increase in union membership increases the probability of a supportive vote by 11.7 ($\pm$3.7) percentage points, while an equally sized decrease in union concentration increases it by 6.5 ($\pm$2.0) points. In contrast, voting in the far less divisive "cash for clunkers" bill (H.R. 3435, passed 316 vs. 109 in favor) is comparatively less related to union influence: an increase in union membership increases the vote probability of a supportive vote by 4.8 ($\pm$2.5) points, while a decrease in union concentration increases it by 4.4 ($\pm$1.9) percentage points.

29. The small number of districts per state (and the existence of at-large districts) would make models including state dummies (instead of random effects) subject to incidental parameters bias.

30. We calculate this difference in probability by increasing the membership variable and respectively decreasing the concentration variable by one standard deviation from the mean while holding all other covariates at their observed value.

CONCLUSION

How does the local organization of membership groups affect their political power to shape law making in the national legislature? Going beyond the literature's often exclusive focus on membership size, we have provided a theoretical rationale for taking into account other organizational features and conducted an empirical investigation of their political effects. Our focus has been on one of the most important membership groups active in democratic politics: labor unions. Drawing on extensive and geographically fine-grained administrative data, we were able to measure two distinct dimensions of local union organization at the level of congressional districts: membership size and the horizontal concentration versus dispersion of union members in different organizational units. Analyzing legislative voting in the US House of Representatives, this is the first study to uncover what we call the concentration effect: for a given level of union membership, representatives from districts where membership is relatively dispersed across multiple unions have more liberal voting records than representatives from districts where members are concentrated. Our analysis demonstrates that the effect of union concentration on legislative voting is not merely an artifact of economic structure, the political power of business, district partisanship, or model assumptions.

Our results have important implications concerning the relevance of labor unions for democratic politics. Previous research has ignored the concentration dimension of local unions and has underestimated the overall political significance of organized labor for national law making. To be clear, we confirm the standard view that group size matters. In addition, however, the district-level concentration of union membership is of nearly equal importance. Our analysis of both DW-NOMINATE scores and individual key votes demonstrates that the two dimensions of local union organization, size and concentration, affect the making of national laws on a broad range of policies. These include policies with large ramifications for the economy and where mass policy preferences are significantly polarized by income. Over the last decade, research has shown that elected representatives are more responsive to high-income groups (e.g., Bartels 2008). Our findings suggest that not just higher overall membership (in line with the state-level analysis of Flavin [2016]) but also lower concentration implies a move toward more political equality. The role of concentration has largely been missing from the debate about union membership decline. While concentration is more or less constant in the period we study, our results suggest that a decrease in concentration may partially offset, up to a point, the political significance of declining membership; in contrast, increasing concentration would augment it.

Furthermore, our findings shed light on different theoretical perspectives on the effect of group organization on political mobilization. A widely held view is that organizational dispersion of group members is a political disadvantage because it increases coordination or collective action problems between group leaders. In contrast, our findings are consistent with the theory of political mobilization based on social interactions in small groups. They suggest that social incentives for political mobilization may be stronger than the presumed perils of a more dispersed membership. This is broadly consistent with a large literature on the significance of social incentives for political behavior. Social incentives are a powerful engine for political action, and they are more pronounced where interactions are less anonymous. One may also conjecture that there is an additional indirect effect. Local group leaders might be more accountable to their members when membership is less concentrated, and so free riding by leaders is reduced.

As a result, we think that, beyond unions, the concentration dimension of group organization makes for a relevant ingredient in more realistic theories of democratic politics. The mechanism underlying our concentration hypothesis is general. Social pressure as a motive for political action is not restricted to unions. Thus, one may expect a similar pattern for, say, churches of the same denomination. Testing this possibility is a clear avenue for further research.

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